Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Digital Signal and Image Processing (01CT1513)	**Aim : Simulate cross correlation and autocorrelation on discrete time signals.**	
Open handed problem -3	**Date:02-08-2026**	**Enrolment No: 92400133079**

Aim :- Simulate cross correlation and

autocorrelation on discrete time signals.

What is Correlation?

Correlation is a method used to determine how similar two signals, functions, or waveforms are to each other. It is commonly used in areas such as signal processing, image processing, and pattern recognition.

There are mainly two types of correlation:
Cross-Correlation
Auto-Correlation

Cross-Correlation
Cross-correlation is used to measure the similarity between two different signals. In this process, one signal is compared with a delayed or shifted version of the other signal.

It helps determine the relationship between two signals and the amount of time delay at which they match most closely.

Cross-correlation is especially useful when a desired signal is mixed with noise or other unwanted signals. It can be used to detect or estimate the original signal from the noisy signal.

Auto-Correlation
Auto-correlation measures the similarity of a signal with a delayed version of the same signal. In other words, a signal is compared with itself at different time delays.

It provides information about the distribution of energy or power within a signal and can also help identify repeating or periodic patterns.

Auto-correlation is commonly used in applications such as radar, sonar, satellite communication, and wireless communication systems.

Devices that use correlation techniques to measure the power of a signal are known as signal correlators.

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Code :

```
import numpy as np
import librosa
import matplotlib.pyplot as plt
from scipy.signal import correlate
import os

original_file = r"D:\SEM 3\python\NYPD [F.A.T.H.E.R EP] - BAGGH-E SMG x FARMAAN SMG x BIG KAY SMG.mp3"
karaoke_file = r"D:\SEM 3\python\NYPD audio.mp3"
different_file = r"D:\SEM 3\python\Bad-Bunny-DtMF-(HipHopKit.com).mp3"

for file in [original_file, karaoke_file, different_file]:
    if not os.path.exists(file):
        print("File not found:", file)
        exit()

sr = 22050

original, _ = librosa.load(original_file, sr=sr, mono=True)
karaoke, _ = librosa.load(karaoke_file, sr=sr, mono=True)
different, _ = librosa.load(different_file, sr=sr, mono=True)

length = min(
    len(original),
    len(karaoke),
    len(different)
)

original = original[:length]
karaoke = karaoke[:length]
different = different[:length]

original = original / np.max(np.abs(original))
karaoke = karaoke / np.max(np.abs(karaoke))
different = different / np.max(np.abs(different))

def normalized_correlation(x, y):
    corr = correlate(x, y, mode="full")

    value = np.max(np.abs(corr)) / (
        np.linalg.norm(x) * np.linalg.norm(y)
    )

    return corr, value

corr1, val1 = normalized_correlation(original, karaoke)
corr2, val2 = normalized_correlation(original, different)
corr3, val3 = normalized_correlation(karaoke, different)

auto_original = correlate(
    original,
    original,
    mode="full"
```

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```
)

auto_karaoke = correlate(
    karaoke,
    karaoke,
    mode="full"
)

auto_different = correlate(
    different,
    different,
    mode="full"
)

plt.figure(figsize=(12, 8))

plt.subplot(3, 1, 1)
plt.plot(original)
plt.title("Original Song - Pal Pal")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.subplot(3, 1, 2)
plt.plot(karaoke)
plt.title("Karaoke Song - Pal Pal")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.subplot(3, 1, 3)
plt.plot(different)
plt.title("Different Song")
plt.xlabel("Samples")
plt.ylabel("Amplitude")

plt.tight_layout()
plt.show()

plt.figure(figsize=(12, 8))

plt.subplot(3, 1, 1)
plt.plot(auto_original)
plt.title("Autocorrelation - Original Song")
plt.xlabel("Lag")
plt.ylabel("Correlation")

plt.subplot(3, 1, 2)
plt.plot(auto_karaoke)
plt.title("Autocorrelation - Karaoke Song")
plt.xlabel("Lag")
plt.ylabel("Correlation")

plt.subplot(3, 1, 3)
plt.plot(auto_different)
plt.title("Autocorrelation - Different Song")
plt.xlabel("Lag")
plt.ylabel("Correlation")
```

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```
plt.tight_layout()
plt.show()

plt.figure(figsize=(12, 8))

plt.subplot(3, 1, 1)
plt.plot(corr1)
plt.title("Cross Correlation - Original vs Karaoke")
plt.xlabel("Lag")
plt.ylabel("Correlation")

plt.subplot(3, 1, 2)
plt.plot(corr2)
plt.title("Cross Correlation - Original vs Different")
plt.xlabel("Lag")
plt.ylabel("Correlation")

plt.subplot(3, 1, 3)
plt.plot(corr3)
plt.title("Cross Correlation - Karaoke vs Different")
plt.xlabel("Lag")
plt.ylabel("Correlation")

plt.tight_layout()
plt.show()

pairs = [
    "Original\nvs\nKaraoke",
    "Original\nvs\nDifferent",
    "Karaoke\nvs\nDifferent"
]

values = [val1, val2, val3]

plt.figure(figsize=(7, 5))

plt.bar(pairs, values)

plt.title("Normalized Correlation Comparison")
plt.ylabel("Correlation Value")

plt.ylim(0, 1)
plt.grid(axis="y")

plt.show()

print("\n==============================")
print("Normalized Correlation Values")
print("==============================")

print(f"Original vs Karaoke   : {val1:.4f}")
print(f"Original vs Different : {val2:.4f}")
print(f"Karaoke vs Different  : {val3:.4f}")

print("\nInterpretation")
```

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```
if val1 > val2 and val1 > val3:
    print("- Original and Karaoke have the highest similarity.")
    print("- This is expected because both are from the same song.")
else:
    print("- Please verify the selected audio files.")

print("\nCorrelation Scale")

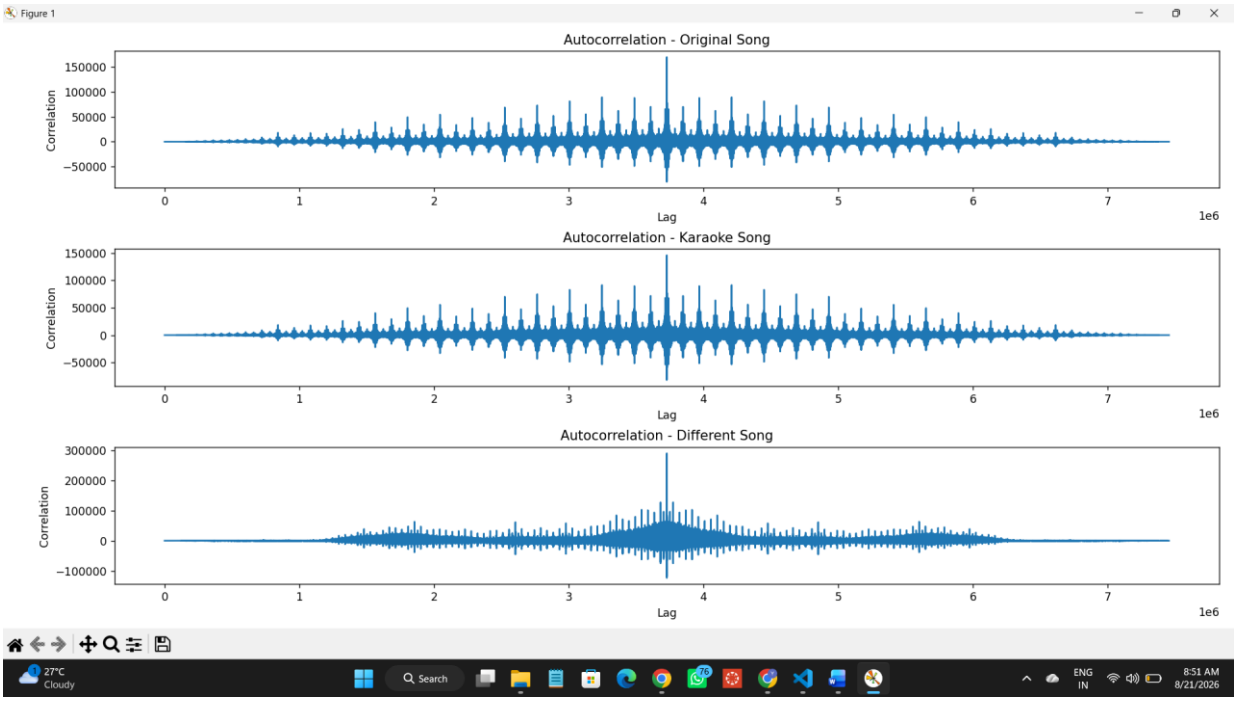
print("0.90 - 1.00 : Very High Similarity")
print("0.70 - 0.89 : High Similarity")
print("0.40 - 0.69 : Moderate Similarity")
print("0.00 - 0.39 : Low Similarity")
```

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Output 1 :

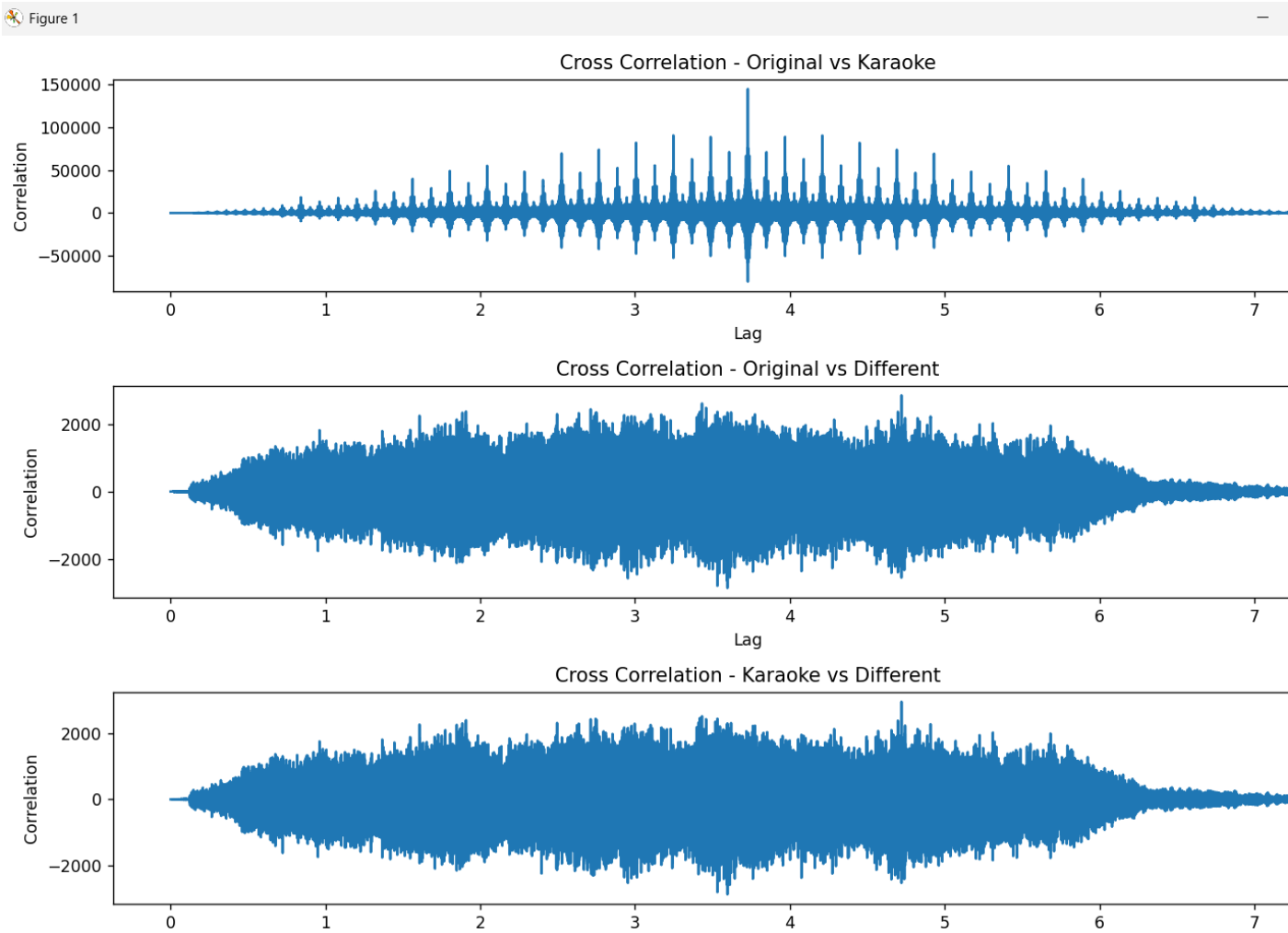


Output 2 : -



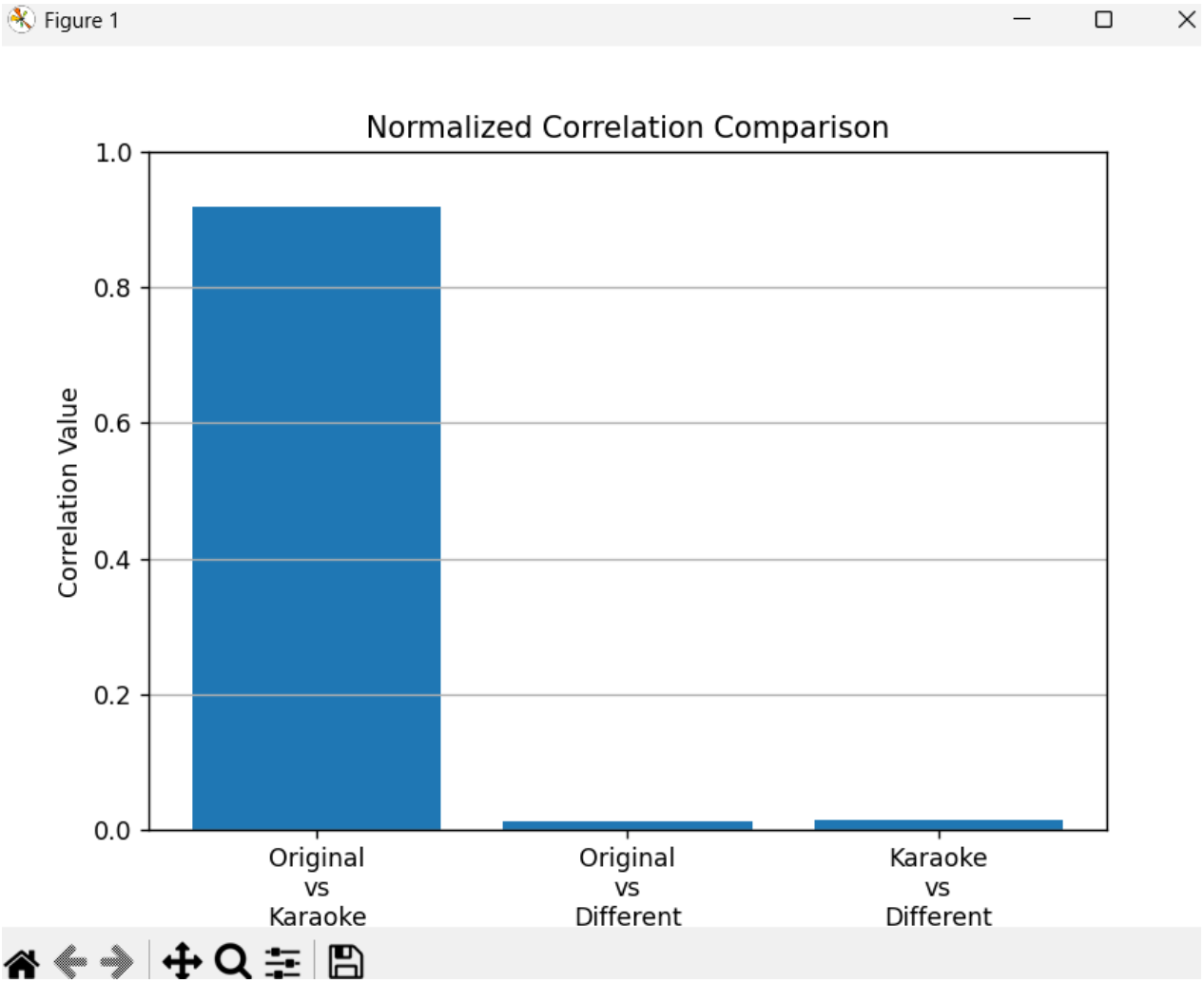
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Output 3 :-



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Output 4:-



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Output 5:-

```
==============================
Normalized Correlation Values
==============================
Original vs Karaoke   : 0.9180
Original vs Different : 0.0128
Karaoke vs Different  : 0.0144

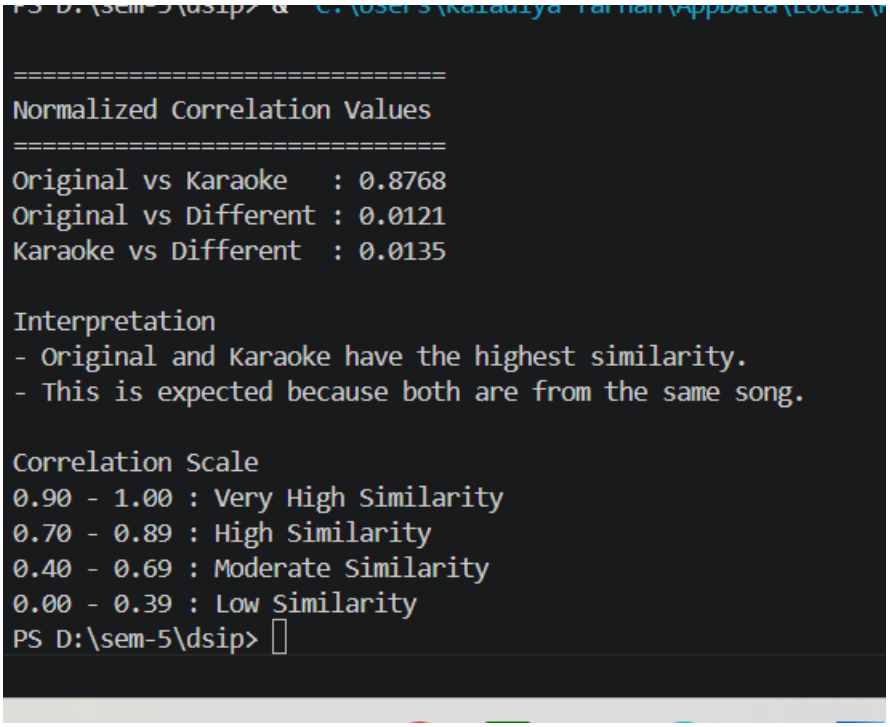
Interpretation
- Original and Karaoke have the highest similarity.
```

Output 6 :

```
Correlation Scale
0.90 - 1.00 : Very High Similarity
0.70 - 0.89 : High Similarity
0.40 - 0.69 : Moderate Similarity
0.00 - 0.39 : Low Similarity
PS D:\SEM 3\python>
```

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Output 6 :-



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1. Conclusion :- This experiment analyzed the similarity between three audio signals using **autocorrelation** and **cross-correlation** techniques.
2. The audio files were first loaded, converted to mono, normalized, and resized to the same length for fair comparison.
3. **Autocorrelation** was used to measure the self-similarity of each audio signal and identify repeating patterns within the same song.
4. **Cross-correlation** was used to compare the similarity between the original song, the karaoke version, and a different song.
5. The waveform plots provided a visual representation of the amplitude variations in each audio signal.
6. The autocorrelation plots showed strong peaks at zero lag, indicating that each signal is highly correlated with itself.
7. The cross-correlation graph between the original song and the karaoke version showed the highest peak, indicating a high degree of similarity.

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8. The cross-correlation values for the original song and the different song were significantly lower, showing that they are not closely related.
9. The normalized correlation values provided a quantitative measure of similarity on a scale from 0 to 1.
10. The bar chart clearly illustrated that the original and karaoke versions had the highest correlation among all comparisons.
11. The different song produced much lower correlation values, confirming that it has a different musical structure and content.
12. Using normalized correlation made the comparison independent of the audio signal amplitudes.
13. The experiment demonstrated that correlation is an effective method for measuring similarity between audio signals.
14. Both waveform visualization and correlation analysis helped in understanding the relationship between different audio tracks.
15. Overall, the results confirm that the karaoke version is most similar to the original song, while the different song has the least similarity, validating the effectiveness of autocorrelation and cross-correlation in digital signal processing applications.